B Guessing strategy and F-score

While F-score is a good metric in some ways, the issue with it is that it incentivizes the model to always guess when it is at least 50% sure that it can get the correct answer. To understand why this is the case, consider the following expression for the F-score:

$$F\text{-score} = \frac{2}{\frac{c+i}{c} + \frac{c+i+n}{c}} = \frac{2c}{2c+2i+n} \ ,$$

where:

- c is the number of correct answers,
- i is the number of incorrect answers, and
- n is the number of non-answered questions.

If you have a greater than $1/2$ chance of being correct, your expected score from guessing is better than the score from not guessing, regardless of the specific values for c, i, and n. This is because the following inequality always holds:

$$\frac{2c+1}{2c+2i+n+2} > \frac{2c}{2c+2i+n+1} \ .$$

The left-hand side represents the expected F-score from guessing, assuming a 50/50 chance of correctness, while the right-hand side is the score from not answering the additional question. Since the denominators are adjusted similarly whether the guess is correct or incorrect, guessing with a probability $> 1/2$ yields a better score.

C Calibration Prompt

```
ANSWER_WITH_CONFIDENCE_PROMPT_TEMPLATE = """
Here is the question:
{question}
Please provide your best guess and a confidence score between
0% to 100% in the following JSON format:
{
    "answer": "Your answer here",
    "confidence_score": number
}
""".strip()
```